

FICMD-Med: A Novel Framework for Imbalanced Class and Modality Distributions of Medical Images

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Abstract—Early disease screening and accurate prediction are vital to modern healthcare. However, single-modality imaging often falls short in capturing the complex, multi-dimensional nature of pathological patterns, thereby limiting diagnostic accuracy and generalizability. To address this, integrating complementary cues from multiple imaging modalities has emerged as a promising approach to enhance predictive robustness and clinical reliability. Nonetheless, real-world multimodal imaging is frequently hindered by class and modality imbalance, which undermines the model’s ability to effectively learn from minority classes and underrepresented modalities. To narrow this gap, we propose FICMD-Med, which integrates a dual-branch architecture for representation enhancement with sample-level gradient optimization. These components are jointly designed to dynamically and adaptively mitigate the coupled adverse effects of class and modality imbalance, particularly under evolving distributions during training, with strong implications for deployment in real-world clinical settings. Extensive experiments on three benchmark datasets validate the effectiveness of FICMD-Med, and demonstrate its superiority over state-of-the-art methods specifically designed for class and modality imbalance; ablation studies further confirm the contribution of each module, particularly the sample-level gradient optimization. Our code is available at <https://github.com/Atlantis-esh/FICMD>.

Index Terms—multimodal, class imbalance, modality imbalance, Prototype-guided contrastive learning, gradient optimization

critical need for early detection and accurate diagnosis in global public health [1].

Early screening and precise prognosis are essential not only for improving patient outcomes and enabling personalized treatment planning, but also for alleviating the overall burden on healthcare systems [4]. However, the initial signs of many severe illnesses are often subtle and heterogeneous, posing significant challenges for automated diagnosis using a single imaging modality such as MRI, CT, PET, or X-ray [5]. For example, structural MRI alone often lacks sufficient sensitivity to detect early-stage AD, potentially delaying timely intervention [6]. In contrast, integrating complementary information from multiple imaging modalities enhances the reliability of identifying individuals at high risk of disease progression. As a result, multimodal fusion has emerged as a promising approach for improving the early and accurate prediction of various diseases [7].

Despite its potential, real-world multimodal medical datasets often suffer from severe distributional imbalances across both modalities and classes. On the modality side, obtaining high-quality, fully paired multimodal imaging data remains costly and logistically challenging, resulting in pronounced modality imbalance [8]. For instance, in COVID-19 diagnosis, X-ray images are far more accessible than CT scans, leading to substantial modality-level sparsity [9]. On the class side, disease severity and pathological states exhibit inherently skewed distributions, with minority classes such as early-stage or rare diseases significantly underrepresented compared to healthy or common conditions. [10] [11]. In brain tumor detection, for instance, tumor-positive cases are significantly outnumbered by healthy samples. Importantly, rare disease samples often suffer from both class and modality imbalance simultaneously. This coupled imbalance results in a systematic loss of critical diagnostic information, introducing challenges that are more severe than those posed by either type of imbalance alone [12].

This dual imbalance presents an essential problem: how to enable unbiased and effective learning from minority classes and underrepresented modalities, without significantly compromising performance on the majority classes and modalities. This problem raises two core challenges: (1) Foundation bias in imbalance-aware strategy design: the coupled effect of class and modality imbalance that introduces bias into representation learning and gradient optimization, which serve as the

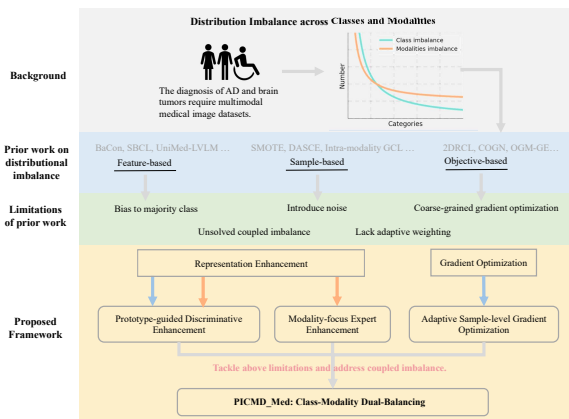


Fig. 1. Motivation of the proposed FICMD-Med framework.

I. INTRODUCTION

The increasing prevalence of high-risk diseases such as Alzheimer’s disease (AD) and cancer has highlighted the

foundation for feature discrimination and optimization dynamics; (2) **Dynamic Adaptation During Training**: the challenge of dynamically adapting to evolving imbalance patterns across both modalities and classes during training, particularly under heterogeneous and incomplete inputs.

Many existing studies address class and modality imbalance independently. For the first challenge, class imbalance is often mitigated through sampling strategies [13], feature-level optimization [14], and related techniques. Concurrently, modality imbalance is typically handled via fusion methods [15] [16], gradient-based optimization [17], and similar approaches. These methods, however, rarely consider the coupled adverse effects of class rarity and modality sparsity. Such coupling can exacerbate weak signals and introduce a fundamental bias into representation learning and gradient optimization, thereby compromising model robustness, particularly for minority classes and underrepresented modalities. For the second challenge, methods like loss reweighting [18], Focal Loss [19], and other static adjustment techniques [20] attempt to correct class imbalance using predefined or heuristic weights. These static schemes fundamentally struggle to adapt to the dynamically evolving imbalance patterns across both modalities and classes during training, especially with heterogeneous and incomplete inputs. Consequently, they often fail to ensure robust model performance.

To tackle these challenges, we propose a unified multimodal learning framework, FICMD-Med, which simultaneously addresses Problems of Imbalanced Class and Modality Distributions in medical images. As shown in Fig. 1, FICMD-Med mitigates class and modality imbalance through two approaches: (1) Representation enhancement, which employs a dual-branch design. The first branch enhances minority-class representations using prototype-guided contrastive learning and a rarity-aware weighting scheme across both classes and modalities. The auxiliary branch introduces dynamic routing to strengthen underrepresented modality-specific expert modeling. (2) Gradient optimization, which adaptively regulates sample-level gradient signals via adaptive optimization, enabling fine-grained learning for underrepresented instances. In summary, our main contributions are as follows:

- We propose a unified framework tailored for complex medical imaging scenarios that simultaneously addresses the challenges posed by imbalanced class and modality distributions. It integrates discriminative representation enhancement with a dual-branch architecture and adaptive gradient optimization, enabling unbiased learning across both modalities and classes.
- Our framework pioneers a jointly adaptive and dynamic mechanism across both classes and modalities. This mechanism strengthens representation learning through prototype-guided contrastive learning with adaptive weighting, and concurrently optimizes gradients by dynamically targeting underrepresented instances at the sample level.
- Experiments conducted on three representative datasets—ADNI, COVID-19 diagnosis and Brain

Tumor classification—demonstrate the superiority of our method over state-of-the-art baselines across multiple metrics, showcasing its robustness and applicability., our method achieves substantial improvements in generalization for real-world medical datasets.

II. RELATED WORK

A. Class Imbalance

Class imbalance often causes models to overfit to majority classes, resulting in biased learning and poor generalization to minority classes [21]. Existing approaches can be broadly categorized into three groups: **Sample-based methods** modify the training distribution through resampling techniques (e.g., SMOTE [13], ADASYN [22]) or data augmentation [23]. However, oversampling may introduce redundant or low-informative samples, while undersampling risks discarding informative data [24]. **Objective-based methods** reshape the loss function to mitigate majority-class dominance, such as Focal Loss [19]. Although straightforward, these methods typically rely on static weighting schemes, limiting their adaptability to evolving training conditions [20]. **Feature-based methods** optimize the embedding space to enhance minority-class representation, often employing contrastive learning to improve class separability [14]. Nevertheless, they remain vulnerable to the influence of majority classes and lack explicit mechanisms to preserve minority feature quality during training [25]. Importantly, most of these methods do not account for the challenges introduced by the co-occurrence of class and modality imbalance, which is an issue commonly encountered in real-world multimodal scenarios.

B. Modality Imbalance

Modality imbalance refers to significant disparities in the availability, quantity, or informativeness of modalities within a multimodal dataset, leading models to overfit dominant modalities and underutilize underrepresented ones [27]. Existing solutions fall into four categories: **Data-level methods** enhance underrepresented modalities via modality-specific augmentation or resampling [15], but incur high computational costs and struggle in high-dimensional settings [27]. **Forward-pass methods** adjust feature processing or fusion strategies to balance modality contributions. While lightweight, they rely on static architectures and struggle with dynamic modality configurations [16]. **Objective-based methods** introduce modality-aware loss weights or targets [18], but often use fixed formulations that fail to capture evolving inter-modal dynamics [28]. **Optimization-based methods** regulate gradients to mitigate the dominance of strong modalities [17], yet assume mild imbalance and perform poorly under severe long-tailed or missing-modality conditions [30]. However, they often treat modality and class imbalance separately, and lack adaptive mechanisms based on joint data distributions.

III. METHODOLOGY

As illustrated in Fig. 2, FICMD-Med integrates three core modules designed to jointly mitigate distributional imbalance

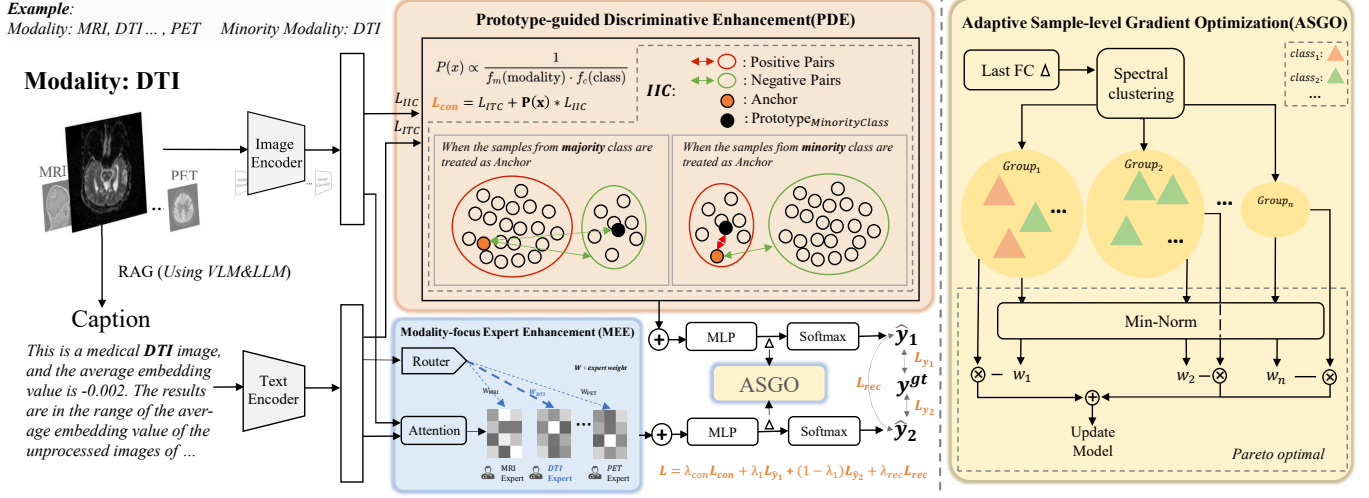


Fig. 2. Overview of the proposed FICMD-Med framework. The framework adopts a dual-branch design for representation enhancement, with the PDE and MEE modules focusing on class-specific and modality-specific aspects, respectively. These two branches jointly address coupled imbalances through targeted loss design. In parallel, the ASGO module performs gradient-level enhancement. This figure illustrates how ASGO conducts sample-level spectral clustering within a training batch. Unlike traditional class-level grouping, this design allows samples—even from different classes—to be clustered based on individual gradient characteristics, enabling fine-grained, instance-aware optimization.

across classes and imaging modalities. Specifically, it employs a dual-branch representation enhancement strategy, consisting of the Prototype-guided Discriminative Enhancement(PDE) and Modality-focus Expert Enhancement(MEE) modules, to enhance feature discriminability for minority classes and underrepresented imaging modalities. In parallel, the Adaptive Sample-level Gradient Optimization(ASGO) module dynamically regulates the training process, enabling more precise handling of sample-level imbalance during optimization.

A. Methodology Review

1) *Problem Definition:* We address the task of disease diagnosis from multimodal medical images. Given a dataset $\mathcal{D} = (x_i, y_i)_{i=1}^N$, where each image x_i is associated with a imaging modality $m_i \in \mathcal{M}$ (e.g., MRI, DTI, PET) and a class label $y_i \in 1, \dots, C$, the data exhibits both class and imaging modality imbalance. Our objective is to develop a robust multimodal classifier $f(x_i) \rightarrow y_i$ that remains effective under this dual-imbalance setting, with a particular emphasis on improving the recognition of minority classes and underrepresented imaging modalities.

2) *Caption Generation Mechanism:* We introduce a lightweight image captioning module to generate imaging modality-aware textual descriptions. For each image x_i , a visual encoder (e.g., ResNet-50 or ViT) extracts high-dimensional features $v_i = E_{\text{img}}(x_i)$. A prompt is then constructed using the imaging modality label m_i and the embedding v_i (e.g., “This is a medical modality image. Average embedding value is v_i . Findings:”) and fed into a medical-specific language model (e.g., BioGPT or DistilBERT) to generate a caption t_i . The generated text serves as a textual modality input and provides auxiliary prototype-level signals. Leveraging textual information can effectively enhance the

representation of medical images, particularly under conditions of limited imaging modalities.

B. Prototype-guided Discriminative Enhancement(PDE)

The PDE Module enhances representation learning by integrating a class prototype mechanism with a supervised image-image contrastive loss ($\mathcal{L}_{IIC}$), weighted by an imbalance-aware strategy. This design encourages intra-class compactness and inter-class separability, with particular benefits for minority classes. Specifically, prototype-based anchoring sharpens decision boundaries between majority and minority classes, while reinforcing the clustering of underrepresented features. The PDE Module is composed of two submodules: IIC and ITC. While the ITC Module facilitates auxiliary unsupervised optimization, our primary focus lies in the IIC Module, which plays a central role in discriminative feature enhancement.

1) *Class Prototype Enhanced Image-to-Image Contrastive Learning:* For each anchor image sample x_i with label y_i and imaging modality m_i , we extract its embedding as $z_i = E_{\text{img}}(x_i)$. The corresponding class prototype is denoted as μ_{y_i} , and the set of all minority-class prototypes is written as $\mu_{\text{minor}} = \{\mu_c \mid c \in \mathcal{C}_{\text{minor}}\}$.

We adopt a supervised contrastive learning formulation, where samples with the same label are treated as positives, and those with different labels as negatives. To address the class imbalance challenge, we introduce a prototype-aware strategy that selectively incorporates class prototypes into the contrastive loss: (i) For majority-class anchors ($y_i \in \mathcal{C}_{\text{major}}$), the anchor is contrasted against all other samples in the batch, and additionally against all minority-class prototypes as negatives. (ii) For minority-class anchors ($y_i \in \mathcal{C}_{\text{minor}}$), the corresponding class prototype μ_{y_i} is also treated as a positive sample, and

all other in-batch samples and minority-class prototypes serve as negatives.

The resulting sample-level contrastive loss is given by:

$$\mathcal{L}_{IIC}(x_i) = -\log \frac{\sum_{z^+ \in \mathcal{P}^+(i) \cup \mathcal{A}_{\text{pos}}(i)} \exp(\text{sim}(z_i, z^+)/\tau)}{\sum_{z \in \mathcal{P}^-(i) \cup \mu_{\text{minor}}} \exp(\text{sim}(z_i, z)/\tau)} \quad (1)$$

where:

- $\mathcal{P}^+(i) = \{z_j \mid y_j = y_i, j \neq i\}$: in-batch positive samples sharing the same label;
- $\mathcal{P}^-(i) = \{z_j \mid j \neq i\}$: all other in-batch samples except the anchor itself;
- $\mathcal{A}_{\text{pos}}(i) = \{\mu_{y_i}\}$ if $y_i \in \mathcal{C}_{\text{minor}}$, otherwise $\emptyset$.

Here, $\text{sim}(\cdot, \cdot)$ denotes cosine similarity and τ is the temperature scaling factor. The final batch-wise loss is obtained by averaging over all samples in the batch: $\mathcal{L}_{IIC} = \frac{1}{B} \sum_{i=1}^B \mathcal{L}_{IIC}(x_i)$, where B represents the batch size. To maintain a representative prototype μ_c for each class c , we dynamically update it during training using an exponential moving average of the feature embeddings computed in each mini-batch. This batch-wise update allows the prototypes to evolve smoothly over time and remain robust to short-term fluctuations in the feature space.

2) *Imbalance-aware Sample Weighting Strategy*: To more effectively capture class and imaging modality imbalance, we assign a sample weight $P(x_i)$ to each sample, defined as follows:

$$P(x_i) \propto \frac{N_{\text{cmax}}}{N_{y_i}} \cdot \frac{N_{\text{mmax}}}{N_{m_i}} \quad (2)$$

Here, N_{y_i} and N_{m_i} denote the number of training samples that share the same class and imaging modality as sample x_i , respectively. N_{cmax} and N_{mmax} represent the maximum number of samples among all classes and imaging modalities. The sample weight $P(x_i)$ controls the contrast strength of samples in IIC, explicitly enhancing the gradient optimization of minority class and imaging modality samples during training.

3) *PDE Module contrastive loss*: To learn consistent cross-modal representations, the PDE module further incorporates the standard Image-Text Contrastive Loss ($\mathcal{L}_{ITC}$), aligning each image with its corresponding RAG-generated text. Since this component follows a conventional formulation and is not the focus of our methodological innovation, we omit detailed discussion for brevity.

Finally, the overall PDE Module loss is:

$$\mathcal{L}_{\text{con}} = \mathcal{L}_{ITC} + P(x_i) \cdot \mathcal{L}_{IIC} \quad (3)$$

The joint embedding vector is obtained by concatenating the text and image representations: $z_i^{\text{PDE}} = \text{Concat}(z_{\text{txt}}, z_{\text{img}})$.

C. Modality-focus Expert Enhancement (MEE)

To address imaging modality imbalance and enhance semantic fusion across heterogeneous medical data, we propose the Modality-Expert Enhancement (MEE) module. At its core, MEE incorporates a dynamic routing mechanism inspired by the Mixture-of-Experts (MoE) paradigm. This

router adaptively assigns imaging modality expert weights based on its distribution, thereby amplifying the influence of underrepresented imaging modalities.

Formally, let $\mathcal{D}_m$ denote the set of training samples for imaging modality m . The modality expert weight w_m for imaging modality m is computed as:

$$w_m = \frac{1/|\mathcal{D}_m|}{\sum_{m'} 1/|\mathcal{D}_{m'}|} \quad (4)$$

This inverse-frequency weighting scheme increases the contribution of rare imaging modalities in the fused representation, effectively mitigating semantic bias induced by data imbalance.

To further align the routing process with imaging modality-specific semantics, the Router is trained using supervised signals derived from ground-truth imaging modality labels. A cross-entropy loss is applied to enforce this alignment constraint. The formula is as follows: $\mathcal{L}_R(\theta_R) = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^K (Y_m)_{ij} \cdot \log(w_m)_{ij}$, where $(Y_m)_{ij}$ is the one-hot encoding of imaging modality labels, $(w_m)_{ij}$ denotes the predicted weight distribution, and θ_R represents the Router parameters.

Supporting this routing mechanism are imaging modality-specific attention heads, which act as domain experts to extract fine-grained semantic features. Given an imaging modality $x_{i,m}$ (e.g., PET, MRI) and its corresponding text description t_i , each attention head independently performs cross-modal attention, using the text embedding ($Q = z_i^{\text{txt}}$) as the query and the image embedding ($K, V = z_{i,m}^{\text{img}}$) as keys and values:

$$\text{Head}_m = \text{Softmax} \left(\frac{(QW_Q^m)(KW_K^m)^\top}{\sqrt{d}} \right) VW_V^m \quad (5)$$

Each attention head captures imaging modality-specific semantics while preserving structural distinctiveness. Finally, all imaging modality-specific attention outputs are aggregated into a unified embedding using the learned fusion weights: $z_i^{\text{MEE}} = \text{Concat}(w_1 \cdot \text{Head}_1, w_2 \cdot \text{Head}_2, \dots, w_M \cdot \text{Head}_M)$

D. Adaptive Sample-level Gradient Optimization(ASGO)

While most existing methods address class imbalance from a class-level perspective, our approach introduces a sample-level optimization strategy. Specifically, FICMD-Med introduce the ASGO Module, which formulates class imbalance mitigation as a multi-objective optimization problem. By clustering per-sample gradients, ASGO adaptively emphasizes minority-class instances during training. Its principal innovation lies in coupling spectral clustering with gradient reconstruction to enable dynamic, context-aware gradient adjustments. To ensure global training stability, class-level objectives are additionally incorporated in a complementary manner.

1) *Sample-Level Spectral Clustering*: The ASGO Module constructs dynamic sub-objectives at the sample level within each training batch. For each sample x_i , it first computes the total loss gradient:

$$\nabla_i = \nabla_{\theta} \mathcal{L}(x_i) \quad (6)$$

To avoid coarse-grained modeling based solely on class labels, the ASGO Module applies spectral clustering to group samples based on the directions of their gradient vectors, forming G gradient-consistent clusters $\{\mathcal{S}_g\}_{g=1}^G$ with corresponding average gradient vectors for group-wise optimization.

Compared to traditional K-means, spectral clustering emphasizes the direction of gradient vectors rather than their magnitude, thereby effectively mitigating the dominance of majority-class samples in gradient updates. This enables more targeted optimization of minority and boundary samples. The sample-level grouping mechanism is a key innovation of this module, enabling the model to dynamically identify and adaptively optimize hard-to-learn boundary and rare samples.

2) *Pareto-Optimal Multi-Objective Solution*: After computing the average gradient for each sample group, the ASGO Module employs a Min-Norm multi-objective optimizer to determine the Pareto-optimal direction within the gradient space of these groups. The objective is to minimize the norm of the weighted sum of group gradients:

$$\min_{\{\alpha_g^{(s)}\}} \left\| \sum_{g=1}^G \alpha_g^{(s)} \nabla_g^{(s)} \right\| \quad \text{s.t.} \quad \sum_{g=1}^G \alpha_g^{(s)} = 1, \quad \alpha_g^{(s)} \geq 0 \quad (7)$$

Here, $\alpha_g^{(s)}$ denotes the optimal weighting coefficient for each sample group, which is used to compute the final sample-level optimization gradient direction.

3) *Joint Strategy*: Building upon the sample-level optimization path, we introduce a tunable class-level Pareto optimization path as an enhancement mechanism to better preserve overall class distribution balance. Specifically, within each training batch, samples are grouped by their class labels y_i , and the average gradient is computed for each class:

$$\nabla_k^{(c)} = \frac{1}{|\mathcal{C}_k|} \sum_{i \in \mathcal{C}_k} \nabla_i \quad (8)$$

Subsequently, class-level weights $\alpha_k^{(c)}$ are computed using the same Pareto optimization strategy, yielding the class-level optimization gradient direction. To integrate fine-grained sample-level control with global class-level balance, the ASGO Module introduces the following joint gradient update strategy:

$$\nabla_{\text{final}} = w_{\text{mix}} \cdot \sum_{g=1}^G \alpha_g^{(s)} \nabla_g^{(s)} + (1 - w_{\text{mix}}) \cdot \sum_{k=1}^C \alpha_k^{(c)} \nabla_k^{(c)} \quad (9)$$

$$\theta \leftarrow \theta - \eta \cdot \nabla_{\text{final}} \quad (10)$$

Here, $w_{\text{mix}} \in [0, 1]$ is the mixture weight which is a hyperparameter, which balances the contributions of sample-level and class-level gradients in the final update. In the experimental section, we conduct an ablation study on different λ values to further validate its regulatory effect.

E. Loss Function

The FICMD-Med framework is optimized using a unified loss function that integrates multimodal fusion, classification, and prediction consistency. It comprises four components: a contrastive loss, classification losses from two prediction branches, and a consistency regularization term.

1) *Supervised Prediction Loss*: The representations produced by the MEE Module and PDE Module, denoted as z_i^{MEE} and z_i^{PDE} , are each passed through their respective multi-layer perceptrons (MLPs) for classification after fusion. The corresponding classification losses are computed using the Cross-Entropy (CE) loss:

$$\hat{y}_1 = \text{Softmax}(\text{MLP}(z_i^{\text{PDE}})), \mathcal{L}_{y_1} = \text{CE}(\hat{y}_1, y_{gt}) \quad (11)$$

$$\hat{y}_2 = \text{Softmax}(\text{MLP}(z_i^{\text{MEE}})), \mathcal{L}_{y_2} = \text{CE}(\hat{y}_2, y_{gt}) \quad (12)$$

Here, $\hat{y}_1$ and $\hat{y}_2$ denote the predicted class probability distributions from the two branches of the model, while y_{gt} represents the ground-truth class label of the medical image.

2) *Consistency Regularization*: To enhance the stability and generalization of the FICMD-Med framework, we introduce a consistency regularization term, $\mathcal{L}_{\text{rec}}$, which enforces alignment between the predicted distributions $\hat{y}_1$ and $\hat{y}_2$ from the MEE Module and the PDE Module branches using symmetric KL divergence. While the MEE Module focuses on semantic modeling via imaging modality-adaptive mechanisms, and PDE Module emphasizes discriminative contrastive learning, both branches classify the same input. Aligning their outputs promotes more stable and generalizable predictions.

The consistency regularization term $\mathcal{L}_{\text{rec}}$ is defined as follows:

$$\mathcal{L}_{\text{rec}} = \frac{1}{2} [D_{\text{KL}}(\hat{y}_1 || \hat{y}_2) + D_{\text{KL}}(\hat{y}_2 || \hat{y}_1)] \quad (13)$$

where $D_{\text{KL}}(p || q)$ denotes the Kullback-Leibler divergence from distribution p to distribution q .

3) *Total Loss Function*: The final loss function of FICMD-Med, denoted as $\mathcal{L}_{\text{total}}$, integrates the aforementioned loss components:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{con}} \mathcal{L}_{\text{con}} + \lambda_1 \mathcal{L}_{y_1} + (1 - \lambda_1) \mathcal{L}_{y_2} + \lambda_{\text{rec}} \mathcal{L}_{\text{rec}}, \quad (14)$$

Here, λ_{con} , λ_{rec} , and $\lambda_1 \in [0, 1]$ are tunable hyperparameters. Specifically, λ_1 modulates the relative contributions of $\mathcal{L}_{y_1}$ and $\mathcal{L}_{y_2}$, enabling adaptive fusion of supervision signals derived from the PDE Module and MEE Module representations.

Minimizing this total loss enables the model to optimize the discriminative power of multimodal features, classification accuracy, and consistency between prediction branches, thereby enhancing overall performance in medical image analysis tasks.

IV. EXPERIMENT

A. Experimental Setups

1) *Datasets*: Following prior work, we evaluate our framework on three multimodal medical datasets. These datasets cover a range of imaging modalities (e.g., MRI, PET, CT) and exhibit varying degrees of class and imaging modality imbalance, as detailed below:

- **Alzheimer's Disease Neuroimaging Initiative (ADNI)¹**: A large-scale neuroimaging dataset comprising structural MRI (sMRI), PET, and DTI scans. We curated a subset

¹<https://adni.loni.usc.edu>

TABLE I
COMPARATIVE PERFORMANCE ACROSS MULTIPLE METRICS ON THE SELECTED ADNI DATASETS.

Method	Acc	CN		MCI		AD		Macro-R	Macro-F1	Macro-AUC
		Recall	F1	Recall	F1	Recall	F1			
Smote [13]	38.45	78.27	53.21	15.37	21	8.04	12.3	33.89	28.84	70.47
DRCL [14]	69.07	76.95	77.78	61.03	59.1	52.1	53.19	63.36	63.36	79.86
SBCL [31]	73.73	86.46	82.09	61.41	63.05	44.91	54.21	64.26	66.45	85.65
GBRIP [32]	75.03	88.15	78.51	57.4	59.44	55.41	71.27	66.69	69.74	87.39
RebLL [33]	77.29	93.2	85.2	55.41	65.12	53.77	70.85	67.46	73.72	90.07
UNIMED-lvlm [5]	78.51	50.92	85.67	50.92	61.65	56.81	66.97	52.88	71.43	89.51
OGM-GE [34]	78.58	100	85.81	48.71	62.45	48.52	64.96	65.74	71.07	88.88
COGN [35]	79.43	96.64	85.37	54.15	65.6	57.89	72.75	69.56	74.57	90.51
AMSS [36]	78.66	99.91	77.3	49.25	60.38	50.99	35.01	66.71	57.56	89.45
FMCD [37]	76.03	98.58	83.49	44.28	59.4	44.99	54.41	62.61	65.77	84.28
Ours	83.04	80.69	87.85	74.05	75.18	64.59	73.4	73.11	78.81	93.3

TABLE II
PERFORMANCE COMPARISON ON COVIDx CT-3A & CXR-4 AND BRAIN TUMOR DATASETS

Method	COVIDx CT-3A & CXR-4			BRAIN TUMOR		
	Acc	Macro-R	Macro-F1	Acc	Macro-R	Macro-F1
Smote [13]	84.51	84.51	85.83	83.58	77.42	84.79
DRCL [14]	87.06	84.31	84.25	52.16	48.57	69.87
SBCL [31]	84.91	83.09	81.77	63.64	62.35	79.10
GBRIP [32]	94.13	93.20	92.11	92.85	91.49	94.03
RebLL [33]	92.49	90.24	91.33	94.60	93.32	92.57
UNIMED-lvlm [5]	90.35	86.55	87.52	50.00	41.85	71.49
OGM-GE [34]	93.74	92.25	92.52	93.59	93.21	93.21
COGN [35]	94.22	91.25	92.57	92.85	90.69	94.68
AMSS [36]	93.00	92.03	91.60	58.80	58.98	76.27
FMCD [37]	93.88	90.65	92.70	58.87	58.38	71.54
Ours	95.35	94.34	93.86	93.69	93.73	95.02

containing 88,508 samples labeled with three cognitive states: Cognitively Normal (CN), Mild Cognitive Impairment (MCI), and Alzheimer’s Disease (AD), with an approximate class distribution of 6:3:1. The imaging modality distribution is similarly imbalanced, approximately 5:2:1, with sMRI as the dominant modality.

- **COVIDx CT-3A & CXR-4**² and **BRAIN TUMOR**³: These two auxiliary datasets focus on classification tasks for pneumonia and brain tumors, respectively. The COVIDx dataset comprises 114,545 CT and X-ray images labeled as COVID-19, Common Pneumonia, or Normal, with a class distribution of approximately 6:3:2 and a highly skewed modality ratio (CT:X-ray $\approx$ 1:17). The BRAIN TUMOR dataset consists of 9,618 CT and MRI scans labeled as Healthy or Tumor, exhibiting a class imbalance of roughly 7:3 and a noticeable modality imbalance in favor of CT.

Furthermore, all three datasets were randomly split into training (80%), validation (10%), and test (10%) sets.

2) *Baseline*: We compare our approach against representative methods from two research directions: class imbalance and imaging modality imbalance. Specifically, for class imbalance,

we consider SMOTE [13], DRCL [14], SBCL [31], GBRIP [32], and RebLL [33]. For imaging modality imbalance, we evaluate UNIMED-LVLM [5], OGM-GE [34], COGN [35], AMSS [36], and FMCD [37].

3) *Evaluation Metrics*: For the ADNI dataset, we adopt Accuracy(Acc) [38], Recall [39], F1 score(F1) [40], and AUC [41]. Accuracy measures the proportion of correctly predicted samples and serves as the most intuitive metric. Recall emphasizes the model’s ability to identify true disease cases, minimizing missed diagnoses, which is core metric in imbalanced medical scenarios due to the high cost of false negatives. The F1 score, defined as the harmonic mean of Precision and Recall, provides a balanced evaluation. AUC assesses the model’s overall discriminative capability and is robust to distribution imbalance. To account for coupled imbalance and better reflect minority-class performance, Recall, F1, and AUC are macro-averaged. For clarity, we denote macro-averaged Recall as Macro-R.

4) *Experiment Setting*: We employ a standard ResNet-50 as the vision encoder in Fig. 2. In Eq.(1), we set τ to 0.07; Four key hyperparameters— w_{mix} in Eq.(9), and λ_1 , $\mathcal{L}_{\text{con}}$, and $\mathcal{L}_{\text{rec}}$ in Eq.(14), are evaluated and analyzed in the subsequent sensitivity analysis. For the proposed framework, we conduct three individual runs with different seeds and report the mean and standard deviation of the results to account for randomness. All models are trained on an NVIDIA RTX 3090 Ti GPU.

B. Comparison with SOTA Baselines

We compare our method with ten state-of-the-art approaches across three public multimodal medical imaging datasets. Table I reports results on ADNI, and Table II summarizes findings on COVIDx CT-3A & CXR-4, and Brain Tumor datasets. On ADNI, our method achieves an accuracy of 83.04%, surpassing the best baseline (COGN, 79.43%) by 3.61%. For minority-class recognition, MCI recall improves by 12.64% over SBCL, and AD recall by 6.7% over COGN. Although CN recall exhibits a slight decline, all other results demonstrate superior performance, leading to an overall well-balanced outcome that particularly benefits minority classes. Our approach also outperforms all compared baselines in both

²<https://www.kaggle.com/datasets/hgunraj/covidxct> and <https://www.kaggle.com/datasets/andyczao/covidx-cxr2>

³<https://www.kaggle.com/datasets/murtozalikhon/brain-tumor-multimodal-image-ct-and-mri>

TABLE III
ABLATION STUDY OF FICMD-MEDI ON THE SELECTED ADNI DATASETS.

Method	Acc	CN		MCI		AD		Macro-R	Macro-F1	AUC
		Recall	F1	Recall	F1	Recall	F1			
FICMD-Med	83.04	80.69	87.85	74.05	75.18	64.59	73.40	73.11	78.81	93.30
Ablation on PDE Module:										
- w/o PDE	77.09	87.27	82.42	65.84	70.09	55.14	70.10	69.42	74.21	88.97
- w/o sample weight in PDE	81.37	94.70	86.03	65.65	73.45	57.26	72.26	72.54	77.25	91.50
- w/o prototype in PDE	81.04	95.57	85.84	62.89	72.42	56.74	71.85	71.73	76.70	91.37
Ablation on MEE Module:										
- w/o MEE	81.98	96.21	86.59	63.88	73.53	58.83	73.62	72.97	77.91	92.30
- w/o expert weight in MEE	82.68	96.27	87.35	66.28	74.62	58.77	73.68	73.77	78.55	92.96
Ablation on ASGO Module:										
- w/o ASGO	78.73	93.61	84.38	69.89	66.64	56.76	67.97	73.42	73.00	88.61

COVID-19 detection and brain tumor diagnosis. These results underscore the robustness and generalizability of our method across diverse medical imaging tasks.

From Table III, we make four key observations: (i) Each module—PDE, MEE, and ASGO—significantly contributes to overall performance. Removing any module results in noticeable performance degradation, confirming the effectiveness of their joint integration and underscoring the importance of both discriminative representation learning and precise gradient optimization. (ii) Under imbalanced class and imaging modality distributions, ASGO and PDE Modules play pivotal roles. (iii) The sample weighting mechanism in the PDE Module is critical. Disabling it (without sample weighting) leads to performance degradation, as the model fails to emphasize minority classes or imaging modalities dynamically. (iv) The prototype-guided mechanism in the PDE Module is also essential. Removing it (without prototypes enhancement) results in inferior performance, suggesting that representation-based class prototypes enhance inter-class boundaries and strengthen the discriminative power of minority classes, with limited impact on the performance of majority classes.

C. Sensitivity Analysis

We conduct a sensitivity analysis on four key hyperparameters in our framework. The detailed experimental settings are described below, and results are visualized in Fig. 3.

Overall, λ_1 and $\mathcal{L}_{con}$ in Eq.(14) are relatively more sensitive compared to the other parameters, whereas variations in $\mathcal{L}_{rec}$ in Eq.(14) and w_{mix} in Eq.(9) have less pronounced effects, indicating the robustness of the framework to those settings. Three key observations emerge: (1) $\lambda_1 = 0.7$ yields the best performance, indicating that discriminative representation learning contributes more significantly to training effectiveness than imaging modality-focus expert-enhanced representations. (2) The optimal value for $\mathcal{L}_{con}$ is 0.3, highlighting its sensitivity. A lower value weakens the model’s ability to enhance minority-class representations, while a higher value tends to suppress majority-class performance (e.g., in the healthy class), leading to bias. Therefore, careful calibration is required. (3) For w_{mix} , the best performance is achieved when it is set to 0.4.

A smaller value results in a sharp performance drop, which validates the importance of the dynamic sample weighting mechanism introduced in ASGO Module.

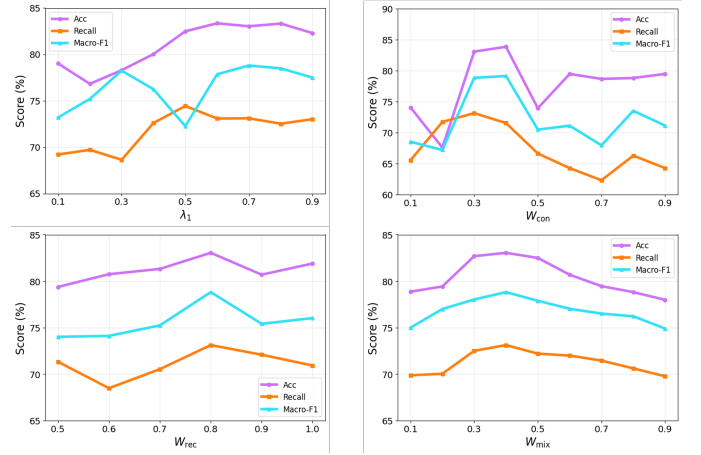


Fig. 3. Performance of FICMD on the selected ADNI dataset with varying four key parameters λ_1 , $\mathcal{L}_{con}$, $\mathcal{L}_{rec}$ and w_{mix} .

V. CONCLUSION

In this work, we propose and validate FICMD-Med, a unified paradigm that addresses both class and imaging modality imbalance in multimodal medical image analysis. By integrating advances in representation optimization, imaging modality-aware interaction, and dynamic gradient regulation, FICMD-Med effectively overcomes the challenges of imbalanced data distributions. Extensive experiments and ablation studies on real-world medical imaging datasets demonstrate the effectiveness, robustness, and scalability of the approach, consistently outperforming state-of-the-art baselines. We anticipate that the insights and methodologies presented herein will provide a solid foundation and practical reference for future research in intelligent medical diagnosis, particularly under complex and imbalanced data scenarios.

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REFERENCES

- [1] Nicholas S. Wilcox, Uri Amit, Jacob B. Reibel, Eva Berlin, Kendyl Howell, and Bonnie Ky, "Cardiovascular disease and cancer: shared risk factors and mechanisms," *Nature Reviews Cardiology*, pp. 617–631, 2024.
- [2] Jifa Zhang, Yinglu Zhang, Jiaxing Wang, Yilin Xia, Jiaxian Zhang, and Lei Chen, "Recent advances in alzheimer's disease: mechanisms, clinical trials and new drug development strategies," *Signal Transduction and Targeted Therapy*, vol. 9, no. 211, 2024.
- [3] Megan C. Roberts, Kathryn E.Holt, Guilherme Del Fiore, Andrea A. Baccarelli, and Caitlin G. Allen, "Precision public health in the era of genomics and big data," *Nature Medicine*, vol. 30, pp. 1865–1873, 2024.
- [4] Rebecca C. Fitzgerald, Antonis C. Antoniou, Ljiljana Fruk, and Nitzan Rosenfeld, "The future of early cancer detection," *Nature Medicine*, vol. 28, pp. 666–677, 2022.
- [5] H. Zhang et al., "Aligning multimodal biomedical images and language via one large vision-language model," in 2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), Dec. 2024, pp. 2874–2879.
- [6] G. Lin et al., "A multimodal MRI-based machine learning framework for classifying cognitive impairment in cerebral small vessel disease," *Science Report*, vol. 15, no. 1, p. 13112, Apr. 2025.
- [7] H. Tan, B. Luo, and C. Cong, "Using deep learning to differentiate between parkinson's disease and multiple system atrophy based on PET and MRI images," in 2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), Dec. 2023, pp. 3181–3187.
- [8] J. Shi, C. Shang, Z. Sun, L. Yu, X. Yang, and Z. Yan, "PASSION: towards effective incomplete multi-modal medical image segmentation with imbalanced missing rates," in Proceedings of the 32nd ACM International Conference on Multimedia, Oct. 2024, pp. 456–465.
- [9] Y. M. G. Costa et al., "COVID-19 detection on chest X-ray and CT scan: a review of the top-100 most cited papers," *Sensors*, vol. 22, no. 19, p. 7303, Sep. 2022.
- [10] W.-S. Choi, H. Lee, D.-S. Han, J. Park, H. Koo, and B.-T. Zhang, "DUEL: duplicate elimination on active memory for self-supervised class-imbalanced learning," in Proceedings of the AAAI Conference on Artificial Intelligence, vol. 38, no. 10, pp. 11579–11587, Mar. 2024.
- [11] S. Rajaraman, Z. Liang, Z. Xue, and S. Antani, "Addressing class imbalance with latent diffusion-based data augmentation for improving disease classification in pediatric chest X-rays," in 2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), Dec. 2024, pp. 5059–5066.
- [12] D. Lepe-Soltero, T. Artières, A. Baudot, and P. Villoutreix, "MODIS: multi-omics data integration for small and unpaired datasets," Mar. 2025, arXiv preprint, arXiv:2503.18856.
- [13] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: synthetic minority over-sampling technique," *Journal Of Artificial Intelligence Research*, vol. 16, pp. 321–357, Jun. 2002.
- [14] C.-L. Duan, Y. Li, X.-S. Wei, and L. Zhao, "Long-tailed object detection pretraining: dynamic rebalancing contrastive learning with dual reconstruction," in The Thirty-eighth Annual Conference on Neural Information Processing Systems, Dec. 2024.
- [15] S. Xu, M. Cui, C. Huang, H. Wang, and D. Hu, "BalanceBenchmark: a survey for multimodal imbalance learning," 2025, arXiv preprint, arXiv:2502.10816.
- [16] Y. Zhou, X. Liang, S. Zheng, H. Xuan, and T. Kumada, "Adaptive mask Co-optimization for modal dependence in multimodal learning," in 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Jun. 2023, pp. 1–5.
- [17] Y. Wei, S. Li, R. Feng, and D. Hu, "Diagnosing and Re-learning for balanced multimodal learning," in European Conference on Computer Vision, 2025, pp. 71–86.
- [18] R. Xu, R. Feng, S.-X. Zhang, and D. Hu, "MMCosine: multi-modal cosine loss towards balanced audio-visual fine-grained learning," in 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Jun. 2023, pp. 1–5.
- [19] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 42, no. 2, pp. 318–327, Feb. 2020.
- [20] Qihao Zhao, Yalun Dai, Shen Lin, Wei Hu, Fan Zhang, and Jun Liu, "LTR: boosting long-tail recognition via reflective learning," in European Conference on Computer Vision, 2024, pp. 1–18.
- [21] S. Wang, R. Liu, K. Li, J. Jiang, and X. Cao, "Class similarity transition: decoupling class similarities and imbalance from generalized few-shot segmentation," in 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), Jun. 2024, pp. 2762–2770.
- [22] H. He, Y. Bai, E. A. Garcia, and S. Li, "ADASYN: adaptive synthetic sampling approach for imbalanced learning," in 2008 IEEE International Joint Conference on Neural Networks, Jun. 2008, pp. 1322–1328.
- [23] M. Qi et al., "DASCE: long-tailed data augmentation based sparse class-correlation exploitation," *IEEE Transactions on Knowledge and Data Engineering*, vol. 37, no. 8, pp. 4497–4511, Aug. 2025.
- [24] M. Peng et al., "Trainable undersampling for class-imbalance learning," in Proceedings of the AAAI Conference on Artificial Intelligence, Association for the Advancement of Artificial Intelligence (AAAI), Jul. 2019, pp. 4707–4714.
- [25] C. Hou, J. Zhang, H. Wang, and T. Zhou, "Subclass-balancing contrastive learning for long-tailed recognition," in Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 5395–5407.
- [26] M. Keramati, L. Meng, and R. D. Evans, "ConR: contrastive regularizer for deep imbalanced regression," presented at the The Twelfth International Conference on Learning Representations, Oct. 2023.
- [27] J. Zhang, G. Liu, Q. Liu, S. Wu, and L. Wang, "Modality-balanced learning for multimedia recommendation," in Proceedings of the 32nd ACM International Conference on Multimedia, in MM '24, 2024, pp. 7551–7560.
- [28] H. Zheng, L. Zhou, H. Li, J. Su, X. Wei, and X. Xu, "BEM: balanced and entropy-based mix for long-tailed semi-supervised learning," presented at the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 22893–22903.
- [29] A. Majee, S. Kothawade, K. Killamsetty, and R. Iyer, "SCoRe: sub-modular combinatorial representation learning," in In Proceedings of the 41st International Conference on Machine Learning, Jun. 2024, pp. 34327–34349.
- [30] Y. Wei and D. Hu, "MMPareto: boosting multimodal learning with innocent unimodal assistance," in Proceedings of the 41st International Conference on Machine Learning, PMLR, Jul. 2024, pp. 52559–52572.
- [31] C. Hou, J. Zhang, H. Wang, and T. Zhou, "Subclass-balancing contrastive learning for long-tailed recognition," in Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 5395–5407.
- [32] J. Huang, Y. Cheung, C. Vong, and W. Qian, "GBRIP: granular ball representation for imbalanced partial label learning," in Proceedings of the AAAI Conference on Artificial Intelligence., 2025, pp. 17431–17439.
- [33] K. Du et al., "Rebalancing multi-label class-incremental learning," in Proceedings of the AAAI Conference on Artificial Intelligence, 2025, pp. 16372–16380.
- [34] X. Peng, Y. Wei, A. Deng, D. Wang, and D. Hu, "Balanced multimodal learning via on-the-fly gradient modulation," in 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Jun. 2022, pp. 8228–8237.
- [35] N. Wu, S. Jastrzebski, K. Cho, and K. J. Geras, "Characterizing and overcoming the greedy nature of learning in multi-modal deep neural networks," Sep. 2022, arXiv preprint, arXiv:2202.05306.
- [36] Y. Yang, H. Pan, Q.-Y. Jiang, Y. Xu, and J. Tang, "Learning to rebalance multi-modal optimization by adaptively masking subnetworks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 47, no. 6, pp. 4553–4566, Jun. 2025.
- [37] Y. Yang, F. Wan, Q.-Y. Jiang, and Y. Xu, "Facilitating multimodal classification via dynamically learning modality gap," presented at the Advances in Neural Information Processing Systems 37 (NeurIPS 2024), 2024, pp. 62108–62122.
- [38] W. Li, F. Lyu, F. Shang, L. Wan, and W. Feng, "Long-tailed learning as multi-objective optimization," in Proceedings of the AAAI Conference on Artificial Intelligence, 2024, pp. 3190–3198.

- [39] F. Diaz, M. D. Ekstrand, and B. Mitra, “Recall, Robustness, and Lexicographic Evaluation,” *ACM transactions on recommender systems*, Apr. 2025.
- [40] N. Thach et al., “Muhboost: multi-label boosting for practical longitudinal human behavior modeling,” in *The Thirteenth International Conference on Learning Representations*, 2025.
- [41] M. B. McDermott, H. Zhang, L. H. Hansen, G. Angelotti, and J. Gallifant, “A closer look at AUROC and AUPRC under class imbalance,” in *Advances in Neural Information Processing Systems 37(NeurlPS 2024)*, 2024, pp. 44102–44163.